

# Tiffany Timbric, Software Developer, Forward Deployed Engineer

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## PROFESSIONAL SUMMARY

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Senior software developer, backend, proficient in Java, JEE, AWS Cloud, Serverless, EC2, S3, Service Mesh, Docker Containerization, Kubernetes, Hexagonal Architecture, microservices, Saga, ETL design, reactive / reactor event processing, messaging (MoM), functional & imperative programming styles, writing readable code, writing secure code, writing safe code.

## TECHNICAL SKILLS

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- **Languages:** Java (25+ yrs), Kotlin, Scala (2+ yrs), Python, Bash, Zsh, Fish Shell
- **Cloud & Infra:** AWS (15+ yrs), Lambdas, S3, SQS, EC2, Service Mesh, Event Bridge, CloudFormation, Kubernetes
- **Frameworks:** Spring Boot, Spring Batch, Reactive Extensions (RxJava), Hibernate, OSGi
- **Data:** PostgreSQL, Oracle, MySQL/MariaDB, Cassandra, MongoDB, Redis (Elasticache)
- **Messaging:** Kafka, SQS, RabbitMQ, ActiveMQ, JMS
- **APIs & Protocols:** Saga, REST, GraphQL, gRPC, Protobuf, WebSockets, HTTP(S), FTP/sFTP
- **Testing:** JUnit, Mockito, WireMock, Cucumber/Gherkin, BDD, TDD
- **Security:** OAuth 2.0/SSO, SSH, HMAC, NTRU, PII protection
- **Dev Tools:** Git, GitHub, GitLab, Bitbucket, Jenkins, Bamboo, Nexus, Artifactory, Atlassian Suite
- **Static Code Analysis Tools:** Sonatype, PMD, CheckStyle, FindBugs
- **Architecture:** Hexagonal Architecture, Microservices, BDD, DDD, Event-Driven, ETL, SOA, PaaS, IaaS, JEE Patterns

## PROFESSIONAL EXPERIENCE

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**Mercor, Micro1.AI** - Denver, CO

*Feb 2026 – (present)*

### Senior Software Engineer, Forward Deployed Engineer

- Reviewed AI generated Spring Boot micro-service and ETL backend code.

**Marriott International** [KMC Contractor] - Denver, CO

*Dec 2024 – Nov 2025*

### Senior Software Engineer — Platform Core

- Designed and built the Universal StarterKit application generator, the standard scaffold for all new platform services:
  - Inbound adapters: REST, Kafka subscriber, GraphQL, gRPC
  - Outbound adapters: REST, streaming REST, Kafka publisher, GraphQL, Elasticache (Redis), MongoDB, PostgreSQL
- Authored the Platform Cache Library supporting Cache-Aside, Read-Through, Write-Around, Write-Back, and Write-Through strategies; LRU/LFU/TTL eviction; L1 in-memory and L2 distributed (Redis) tiers.
- Built a Streaming REST Client Platform Library enabling unlimited-size transfers using Reactive Extensions, Spring Data (JPA, r2dbc), and FasterXML Jackson — adopted across multiple product teams.
- Authored the team's Hexagonal Architecture Sample Application, establishing reference patterns for the platform.
- Stack: Java 21, Spring Boot, AWS, S3, Kubernetes, Maven, JUnit, Lombok, Immutables.

**Amtrak** [Allwyn Contractor] - Denver, CO

*Feb 2023 – Apr 2024*

### Senior Software Engineer

- Sole author (30,000+ lines) of a full AWS rewrite of Amtrak's customer-facing outbound email system, covering purchase confirmations and trip itineraries across all stations, passenger classes, add-ons, and third-party carriers.
- Designed observability into the system from the ground up: business and system metrics, alerting, and structured logging with PII protection at all log levels.
- Stack: Java 11, AWS (CloudFormation, Lambda, Event Bridge, SQS, S3), Spring, FasterXML Jackson, Maven, Immutables.

**GHX** [Ascendion Contractor] - Louisville, CO

*Jun 2022 – Jan 2023*

### Senior Software Engineer

- Led a team building integration middleware connecting hospital ERP and medical equipment procurement systems to the GHX exchange platform.
- Managed bi-weekly production release cadence with deployments into hundreds of instances.
- Stack: Java, AWS (EC2, SQS, S3, Lambda), Oracle, REST, JSON, FTP/sFTP.

**PayPal** [TCS Contractor] - Denver, CO

*Feb 2020 – May 2022*

#### Software Engineer

- Implemented direct deposit signup services in Spring Boot and AWS.
- Built Spring Batch pipeline to migrate bank account data across all PayPal accounts during bank-change operations.
- Delivered enhancements to the merchant batch report generation system; provided L3 application support.

**Arris / CommScope** - Denver, CO

*Jan 2019 – Oct 2019*

#### Software Engineer

- Modernized the video catalog and merchandising backend (originally authored by Motorola) for Bouygues Telecom, France.
- Stack: Java EE, Wildfly, Glassfish, WebSphere, REST, WireMock, Mockito, Postman, Wireshark.

**Comcast** [Arris Contractor] - Denver, CO

*Jan 2018 – Apr 2018*

#### Software Engineer

- Migrated a Java monolith to Docker/Kubernetes on AWS; upgraded to Java 11, Angular, Reactive Extensions, and Cucumber.

**Charter Communications** [Appzlogic Contractor] - Denver, CO

*Jan 2016 – Dec 2017*

#### Software Engineer

- Maintained and extended 100+ Java microservices on AWS, leveraging Netflix OSS, Cassandra, RxJava, SQS, SNS, S3, and ActiveMQ.

**Comcast** [Contractor] - Denver, CO

*Sep 2014 – Feb 2015*

#### Software Engineer

- Designed and built RxJDBC (R2BC) — a Reactive Extensions + RabbitMQ front-end to Oracle DB as part of the platform Resiliency Project.

**Tendril, Inc.** - Boulder, CO

*Feb 2014 – May 2014*

#### Software Engineer

- Back-end development on SOA microservices using Finagle, Protobuf, Scala, and cloud infrastructure; BDD with Cucumber/PostgreSQL.

**JDA Software Group, Inc.** - Denver, CO

*Feb 2010 – Dec 2013*

#### Senior Software Engineer

- Project lead integrating iOS/Android mobile task management with legacy J2EE web platform, serving Fortune 500 customers.
- Engineered for scalability and performance across high-volume enterprise deployments.
- Stack: OSGi, Gradle, Ext-JS, SQL Server, AWS, Git/Mercurial/SVN.

**Independent Consultant** - Denver, CO

*Jan 2008 – Dec 2009*

#### Software Developer

- Extended and maintained a JEE merchandise inventory system for clients including BellSouth, eBay, Frontier Airlines, AmDocs, and E-tilize. Stack: Java, Spring, MySQL.

**Evolution Hosting, Inc.** - Louisville, CO

*Feb 2007 – Dec 2007*

#### Chief Operating Officer

- Operated Linux-based data center hosting J2EE applications for clients including Wells Fargo, Crocs, and Hunter Douglas.
- Stack: J2EE, Linux, VMware, Xen, AWS, MySQL.

**Independent Consultant** - Denver, CO

*Feb 2004 – Jan 2007*

#### Software Developer

- Built a high-throughput NIO client achieving 100 Mb/s message rates.

- Consulted at eBay, Frontier Airlines, BellSouth, and Skyway representing Sun Microsystems.

**Packexpo, Inc.** - Fairfax, VA

*Feb 2002 – Feb 2004*

#### **Software Architect**

- Designed and built a reporting system for packaging advertising services. Stack: JBoss, Linux, MySQL, Java, Groovy, JSP.

**Sun Microsystems, Inc.** - Broomfield, CO

*Feb 1999 – Jan 2002*

#### **Java Consultant**

- Represented Sun delivering RUP/OOA&D mentoring and J2EE architecture assessments at Citigroup, Vanguard Fiduciary Trust, and Credit Suisse — recommended SOA/JEE modernization that was subsequently adopted.

**Sun Microsystems, Inc.** - Broomfield, CO

*Apr 1998 – Jan 1999*

#### **Java Developer**

- Built web-based and Swing computer-assisted training systems; improved a custom parser template engine via result caching and algorithm optimization.

### **ARCHITECTURE & CODE SAMPLES**

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- Enterprise Architecture & Modeling: [sparxsystems.com](http://sparxsystems.com)
- Finite State Machine (Java): <https://github.com/chadwickboggs/fsm>
- Cryptography (Java): <https://github.com/chadwickboggs/crypto>
- Spring Boot Sample: <https://github.com/chadwickboggs/xchange>
- Design Pattern: Hexagonal Architecture — In Adapters → In Ports → Service → Out Ports → Out Adapters

### **EDUCATION, CERTIFICATIONS & PATENTS**

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- Bachelor of Science, Computer Science — University of Colorado, 1998
- Electrical Engineering — U.S.M.A. at West Point (attended)
- Sun Certified Java Developer (SCJD) | Sun Certified Java Programmer (SCJP)
- AWS Certified Developer — Associate (in progress)
- U.S. Patent #6,907,423 — Search Engine Interface and Method of Controlling Client Searches
- Training: Scala (courses by Martin Odersky, 2+ yrs), Kotlin (2+ yrs), Python (2+ yrs)